

NACA 4609 Aerodynamic Analysis: Thin Airfoil Theory, Joukowski Mapping, and Navier–Stokes CFD

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Abstract

This study investigates the aerodynamic characteristics of the cambered NACA 4609 airfoil, comparing and contrasting classical inviscid theories with viscous Navier-Stokes CFD simulations. Baseline data was generated by developing a custom MATLAB algorithm to automate the computation of aerodynamic force coefficients and surface pressure distributions using Thin Airfoil Theory and the Joukowski Conformal Mapping technique. These classical inviscid baselines were evaluated against a $\approx 106,000$ -element, unstructured C-grid computational fluid dynamics (CFD) simulation utilizing the $k - \omega$ Shear Stress Transport (SST) turbulence model across an angle of attack sweep from 0° to 20° . Results demonstrate that within the attached flow regime ($0^\circ \leq \alpha \leq 10^\circ$), classical inviscid methods provide highly accurate approximations of lift slope and pressure recovery. However, as the angle of attack increases, the classical models fail to capture viscous effects and adverse pressure gradients, mathematically predicting the growth of lift beyond the physical stall regime. The viscous CFD simulation successfully captures the onset of trailing-edge separation at $\alpha = 15^\circ$ and deep stall at $\alpha = 20^\circ$. Ultimately, this comparative investigation highlights the relevance and utility of classical inviscid theories for rapid conceptual design, while explicitly defining the limits where full viscous modeling becomes necessary.

1 Introduction

A fundamental challenge throughout the development of fluid dynamics has been the accurate prediction of aerodynamic forces for lifting bodies such as airfoils. The NACA 4609 is a cambered airfoil characterized by its high camber of 4% and its relatively thin profile of 9%. Cambered airfoils are advantageous since they naturally produce substantial lift when compared to symmetric airfoils, making them suitable for high-lift requirements and turbomachinery applications.

Historically, prior to the advent of advanced computers and numerical simulations, engineers relied on analytical methods to compute the necessary aerodynamic force coefficients. While being mathematically elegant, these methods weren't without limitations.

Techniques such as Thin Airfoil Theory and the Joukowski transformation operate under the assumption of inviscid flow. Consequently, they are unable to capture the boundary layer and viscous effects, thereby completely ignoring important phenomena such as flow separation.

However, with the exponential increase in computational power, solving the full Navier-Stokes equations is now viable and has become very effective for simulating complex aerodynamic flows. The objective of this study is to rigorously evaluate, compare, and contrast the aerodynamic characteristics of the NACA 4609 across a sweep of angles of attack ranging from 0° to 20° . By comparing theoretical baseline data—generated via algorithmic implementations of Thin Airfoil Theory and Conformal Mapping in MATLAB—against a viscous, steady-state computational fluid dynamics (CFD) simulation utilizing the $k-\omega$ SST turbulence model, this study aims to explicitly demonstrate the limitations of inviscid theory and highlight where viscous modeling becomes necessary.

2 Theoretical Framework

In order to fully grasp the CFD results obtained using the complete viscous solution, baseline data was obtained using two classical methodologies that have been used throughout the last century. Namely, Thin Airfoil Theory (Small Perturbations) and Conformal Mapping technique via the Joukowski Transform. Since these methods are inherently analytical, A custom MATLAB code along with a Graphical User Interface (GUI) was written to automate the process of obtaining the coefficients (including numerical integrations) and generalize the results for different geometries should one desire to simulate other airfoils.

2.1 Thin Airfoil Theory (TAT)

The first method we shall examine is the Thin Airfoil Theory abbreviated TAT for simplicity. In TAT, the airfoil is assumed to be infinitely thin, in other words, with no perceived thickness. It then simulates the flow by having vortex sheet placed along the camber line, but since our airfoil is very thin, it approximately falls on the chord. The objective then becomes computing $\gamma(x)$, so that the camber line essentially becomes a streamline of the flow itself and the Kutta condition $\gamma(TE) = 0$ is satisfied. The NACA 4609 possesses a maximum camber of $m = 0.04$ located at $p = 0.6$ chord lengths from the leading edge. The mean camber line, $z_c(x)$, is defined piecewise as:

$$z_c(x) = \begin{cases} \frac{m}{p^2} \left[2p \left(\frac{x}{c} \right) - \left(\frac{x}{c} \right)^2 \right], & 0 \leq \frac{x}{c} \leq p \\ \frac{m}{(1-p)^2} \left[(1-2p) + 2p \left(\frac{x}{c} \right) - \left(\frac{x}{c} \right)^2 \right], & p \leq \frac{x}{c} \leq 1 \end{cases} \quad (1)$$

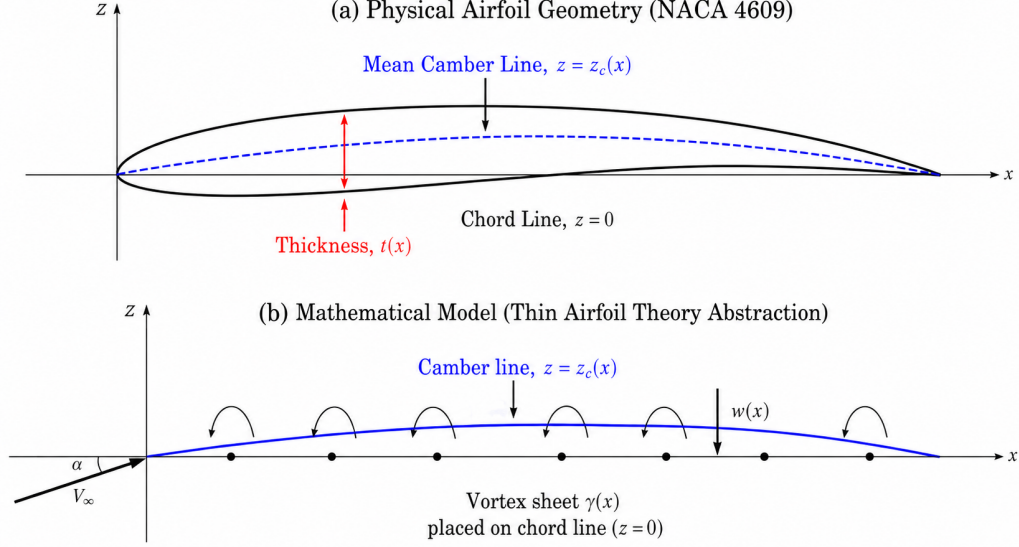


Figure 1: Geometric and mathematical visualization of TAT for the NACA 4609. (a) The physical geometry of the airfoil with the mean camber line $z_c(x)$ are shown. (b) TAT visualization with the vortex sheet is shown while the induced downwash $w(x)$ ensures the flow tangency along the camber line.

The fundamental equation of Thin Airfoil Theory enforces the flow tangency boundary condition along the camber line, equating the induced velocity from the vortex sheet, $\gamma(\xi)$, to the freestream normal velocity: (note that ξ is just a dummy variable used to ease the process of integration, it's an analog of x that varies from 0 to c , where x corresponds to the chord c .)

$$V_\infty \left(\alpha - \frac{dz_c}{dx} \right) = \frac{1}{2\pi} \int_0^c \frac{\gamma(\xi)}{x - \xi} d\xi \quad (2)$$

To evaluate the aerodynamic integrals along the chord, a coordinate transformation is applied to map x to θ :

$$\xi = \frac{c}{2}(1 - \cos \theta) \quad \implies \quad d\xi = \frac{c}{2} \sin \theta d\theta \quad (3)$$

And since x is fixed, it correlates with a particular value of θ , namely θ_0 :

$$x = \frac{c}{2}(1 - \cos \theta_0) \quad \implies \quad dx = \frac{c}{2} \sin \theta_0 d\theta_0 \quad (4)$$

To solve the above integral equation (2), a Fourier series expansion is introduced for the vortex sheet strength $\gamma(\theta)$: (Refer to [1] for full derivation)

$$\gamma(\theta) = 2V_\infty \left(A_0 \frac{1 + \cos \theta}{\sin \theta} + \sum_{n=1}^{\infty} A_n \sin(n\theta) \right) \quad (5)$$

Substituting this expansion into the fundamental equation yields the integrals for the Fourier coefficients A_0 and A_n . These integrals evaluate the derivative of the camber line strictly as a function of θ , $\frac{dz_c(\theta)}{dx}$:

$$A_0 = \alpha - \frac{1}{\pi} \int_0^\pi \frac{dz_c(\theta_0)}{dx} d\theta_0 \quad (6)$$

$$A_n = \frac{2}{\pi} \int_0^\pi \frac{dz_c(\theta_0)}{dx} \cos(n\theta_0) d\theta_0 \quad \text{for } n = 1, 2 \quad (7)$$

Once the custom MATLAB algorithm numerically computes these Fourier coefficients for our specific airfoil which is a NACA 4609, then the total circulation (Γ) around the airfoil can be found by integrating the vortex sheet along the chord c . Applying the transformation $d\xi = \frac{c}{2} \sin \theta d\theta$ yields:

$$\Gamma = \int_0^c \gamma(\xi) d\xi = \int_0^\pi \gamma(\theta) \frac{c}{2} \sin \theta d\theta \quad (8)$$

Substituting the Fourier expansion of $\gamma(\theta)$ into the integral, the circulation simplifies to:

$$\Gamma = cV_\infty \pi \left(A_0 + \frac{A_1}{2} \right) \quad (9)$$

Applying the Kutta-Joukowski theorem ($L' = \rho V_\infty \Gamma$), the lift per unit span is calculated. Non-dimensionalizing the lift per unit span using the freestream dynamic pressure q defined as $\frac{1}{2} \rho V_\infty^2 c$ yields the final analytical expression for the Thin Airfoil Theory lift coefficient:

$$C_l = \frac{L'}{\frac{1}{2} \rho V_\infty^2 c} = \frac{\rho V_\infty [cV_\infty \pi (A_0 + \frac{A_1}{2})]}{\frac{1}{2} \rho V_\infty^2 c} = \pi(2A_0 + A_1) \quad (10)$$

In a similar manner, the pitching moment coefficient about the leading edge is derived by integrating the moment generated by each differential vortex element ($dM = -x \cdot dL$), yielding: (following the aerodynamic sign convention where pitching down (counterclockwise) is taken to be negative)

$$C_{m,LE} = -\frac{\pi}{2} \left(A_0 + A_1 - \frac{A_2}{2} \right) \quad (11)$$

2.2 Conformal Mapping and Geometric Approximation

Conformal Mapping technique is one of the most elegant and mathematically imaginative methods to simulate airfoils. Unlike TAT, where the entire airfoil is reduced to its camber line, the Conformal Mapping method accounts for the thickness by using a special transform called the Joukowski transform to map flow over a circle in the complex plane ζ to an airfoil shape in the z plane. (Since flow over a simple circle is much easier to obtain analytically compared to a complicated airfoil.) The transformation is governed by:

$$z = \zeta + \frac{a^2}{\zeta} \quad (12)$$

where a is a real constant defining the mapping scale, and the resulting chord length is approximated as $c \approx 4a$.

However, this method isn't without its limitations; The family of airfoils produced by this method are often called "Joukowski Airfoils". Any one unique Joukowski airfoil is produced by setting specific dimensions for the circle, namely Radius and Center coordinates. Consequently, one has varying control only over these parameters. Therefore, there isn't a one-on-one correspondence with any arbitrary airfoil geometry like the

NACA family. In order to approximate the specific circle dimensions for our NACA 4609 in the ζ plane, the following approximations and shifts were made;

For a cambered, thick airfoil, the center of the circle in the ζ -plane cannot be placed at the origin. Instead, we must shift it by a complex offset $\zeta_c = -\epsilon + ih$. Also, in order to satisfy the Kutta condition and the fact that the flow leaves the trailing edge smoothly, the circle's perimeter ought to pass through the singular point $\zeta = a$. Therefore, our circle's radius, R , is subject to the following constraint:

$$R = \sqrt{(a + \epsilon)^2 + h^2} \quad (13)$$

The real shift ϵ dictates the airfoil's thickness, while the imaginary shift h dictates its camber. To generate the closest match for our NACA 4609 airfoil or any other arbitrary 4-digit NACA airfoil, the custom MATLAB algorithm applies an approximation to match the maximum thickness-to-chord ratio ($t/c = 0.09$) and the maximum camber-to-chord ratio ($m/c = 0.04$) using the fundamental Joukowski relations:

$$\frac{t}{c} \approx \frac{3\sqrt{3}}{4} \left(\frac{\epsilon}{a} \right) \quad (14)$$

$$\frac{m}{c} \approx \frac{1}{2} \left(\frac{h}{a} \right) \quad (15)$$

By solving these equations, the algorithm computed the required ϵ and h coordinates to offset the circle, applying the transformation to yield the closest possible Joukowski equivalent to the NACA 4609. ("The airfoil equivalency approximations, specifically the maximum thickness and camber approximations, were derived utilizing first-order Taylor series expansions of the complex mapping function, details of which can be found in [5, 6].")

However, it is necessary to point out the geometric discrepancy that exists in this approach; You see, the Joukowski transformation naturally fixes the location of maximum camber at 50% of the chord. This is due to the inherent geometric symmetry of the circle. While our NACA 4609 has its maximum camber at 60% of the chord. Furthermore, this transformation is mathematically required to have a cusped (i.e sharp) trailing edge where in reality the NACA 4609 features a trailing edge with a finite thickness. Despite these minor geometric discrepancies, the Conformal Mapping approach as we shall see served as an excellent baseline for our comparison and showed good agreement with the full CFD solution.

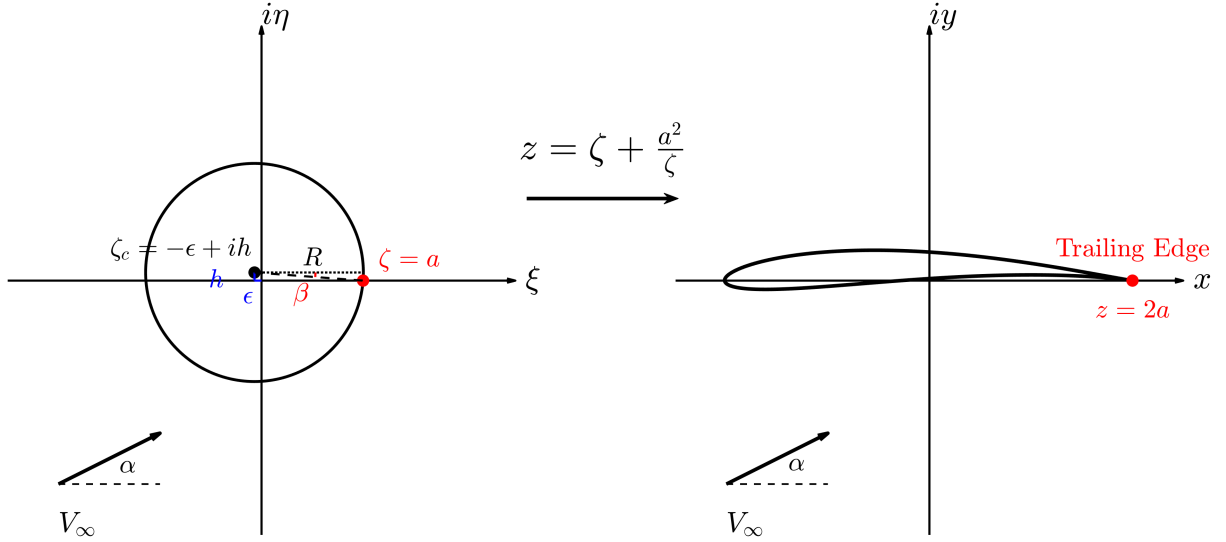


Figure 2: Visual representation of the Joukowski Conformal Mapping technique, demonstrating the transformation of a rotated, offset circle in the complex ζ -plane into a cambered airfoil in the physical z -plane. The center offset μ (mathematically represented as $\zeta_c = -\epsilon + ih$) dictates the shift required to closely approximate the dimensions associated with the NACA 4609, namely thickness and camber.

2.2.1 Complex Potential and the Kutta Condition

To compute the aerodynamic lift generated by the mapped airfoil, the flow must first be defined in the complex ζ -plane. The fundamental flow over the offset, rotating circle is governed by the complex potential function, $W(\zeta)$. This function superimposes a uniform freestream velocity (V_∞) approaching at an angle of attack (α), a doublet to simulate the solid circle, and a free vortex to simulate the circulation (Γ).

For a circle of radius R centered at $\zeta_c = -\epsilon + ih$, the complex potential is defined as:

$$W(\zeta) = V_\infty \left[(\zeta - \zeta_c) e^{-i\alpha} + \frac{R^2}{\zeta - \zeta_c} e^{i\alpha} \right] + \frac{i\Gamma}{2\pi} \ln(\zeta - \zeta_c) \quad (16)$$

The complex velocity field, $w(\zeta)$, is obtained by taking the derivative of the potential function with respect to ζ :

$$w(\zeta) = \frac{dW}{d\zeta} = V_\infty \left[e^{-i\alpha} - \frac{R^2}{(\zeta - \zeta_c)^2} e^{i\alpha} \right] + \frac{i\Gamma}{2\pi(\zeta - \zeta_c)} \quad (17)$$

To translate this ideal flow into physical aerodynamic forces, the Kutta condition must be applied. In physical reality, fluid cannot sustain infinite velocity around a sharp corner. However, the derivative of the mapping function, $\frac{dz}{d\zeta} = 1 - \frac{a^2}{\zeta^2}$, evaluates to zero exactly at the trailing edge coordinate $\zeta = a$. The velocity in the physical z -plane is then given by the chain rule:

$$\frac{dW}{dz} = \frac{dW/d\zeta}{dz/d\zeta} \quad (18)$$

To prevent the physical velocity ($\frac{dW}{dz}$) from becoming infinite at the trailing edge, the complex velocity on the circle ($\frac{dW}{d\zeta}$) must simultaneously equal zero at $\zeta = a$. This forces the rear stagnation point of the circle to coincide with the singular point that exists.

Setting $w(a) = 0$ requires knowing the geometric distance from the offset center to the trailing edge. From the geometry, $a - \zeta_c$ can be expressed in polar form using the offset angle β :

$$a - \zeta_c = (a + \epsilon) - ih = Re^{-i\beta} \quad (19)$$

Substituting $\zeta = a$ and $a - \zeta_c = Re^{-i\beta}$ into the complex velocity equation yields:

$$0 = V_\infty \left[e^{-i\alpha} - \frac{R^2}{(Re^{-i\beta})^2} e^{i\alpha} \right] + \frac{i\Gamma}{2\pi Re^{-i\beta}} \quad (20)$$

Simplifying the exponents and multiplying the entire equation by $e^{-i\beta}$:

$$0 = V_\infty [e^{-i(\alpha+\beta)} - e^{i(\alpha+\beta)}] + \frac{i\Gamma}{2\pi R} \quad (21)$$

Using Euler's identity for the sine function ($e^{-ix} - e^{ix} = -2i \sin x$), the equation reduces to:

$$0 = -2iV_\infty \sin(\alpha + \beta) + \frac{i\Gamma}{2\pi R} \quad (22)$$

Solving this directly for the circulation (Γ) yields the exact vortex strength required to satisfy the Kutta condition:

$$\Gamma = 4\pi RV_\infty \sin(\alpha + \beta) \quad (23)$$

2.2.2 Lift Calculation

Now that the circulation is established, we proceed to compute actual lift. According to the Kutta-Joukowski theorem, the lift per unit span (L') is equal to:

$$L' = \rho V_\infty \Gamma \quad (24)$$

Non-dimensionalizing this force by the freestream dynamic pressure ($q = \frac{1}{2}\rho V_\infty^2$) and the approximated chord length ($c \approx 4a$) yields the analytical expression for the lift coefficient:

$$C_l = \frac{L'}{\frac{1}{2}\rho V_\infty^2 c} = \frac{\rho V_\infty [4\pi RV_\infty \sin(\alpha + \beta)]}{\frac{1}{2}\rho V_\infty^2 c} = \frac{8\pi R}{c} \sin(\alpha + \beta) \quad (25)$$

$$\beta = \tan^{-1} \left(\frac{h}{a + \epsilon} \right) \quad (26)$$

Based on the extracted parameters for the NACA 4609 geometry (where $h = 0.08$ and $\epsilon \approx 0.0693$), the offset angle was computed as $\beta \approx 4.28^\circ$ and Radius as $R \approx 1.0723$. Also, a was taken as 1 for the sake of mathematical simplicity.

Straight away, the above lift derivation demonstrates that the lift coefficient varies linearly with the sine of the angle of attack. Essentially, predicting an infinitely rising linear lift slope without capturing real world phenomenon such as flow separation induced by viscous effects and boundary layers.

2.3 Algorithmic Implementation and Numerical Solver

To automate and speed up the process of coefficient calculations across a relatively wide range of angles of attack ($0^\circ \leq \alpha \leq 20^\circ$) and to further ensure repeatability and to eliminate manual errors, a custom MATLAB solver was developed for this very task. Furthermore, the algorithm was generalized to be able to obtain coefficients for any arbitrary NACA 4-digit airfoil via a custom Graphical User Interface (GUI), should the user wish to do so.

For the Thin Airfoil Theory analysis, the piecewise mathematical definition of the NACA 4609 camber line $z_c(x)$ was discretized into a large one-dimensional array along the angular coordinate domain $0 \leq \theta \leq \pi$. The derivative $\frac{dz_c(\theta)}{dx}$ was computed computationally, and the A_0 , A_1 , and A_2 coefficients were computed using numerical integration functions to output the linear lift slope.

Simultaneously, a separate algorithm was developed to handle the Joukowski Conformal Mapping implementation. By simply inputting the target thickness-to-chord ($t/c = 0.09$) and camber-to-chord ($m/c = 0.04$) ratios, the solver automated the calculation of the geometric circle offsets (ϵ and h) and mapped the complex ζ -plane coordinates to the physical z -plane. It is noteworthy to mention, that this algorithm was also generalized to compute the offset and lift calculations for any arbitrary NACA 4-digit airfoil in a similar manner to TAT.

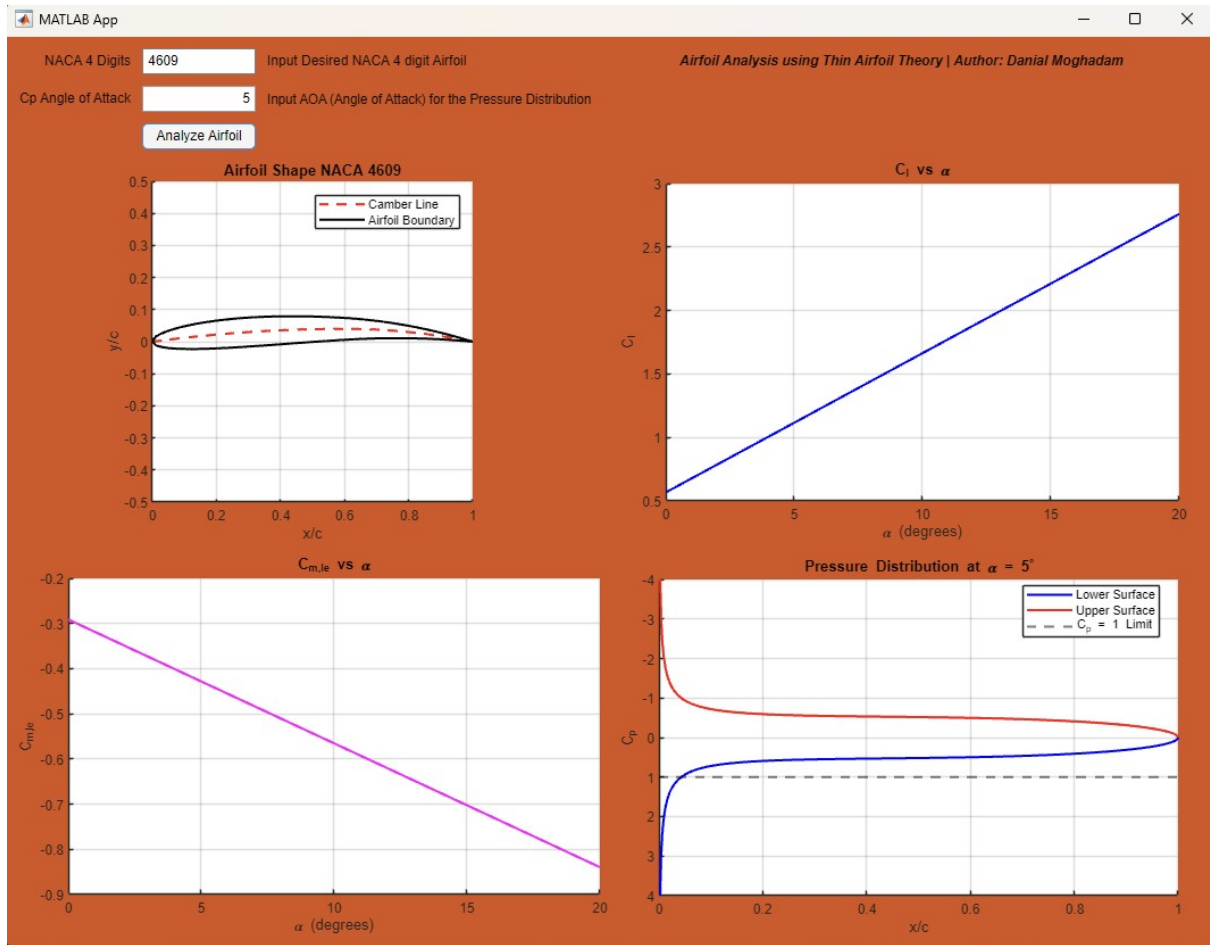


Figure 3: A snapshot of the custom Graphical User Interface (GUI) developed in MATLAB to automate the Thin Airfoil Theory computations.

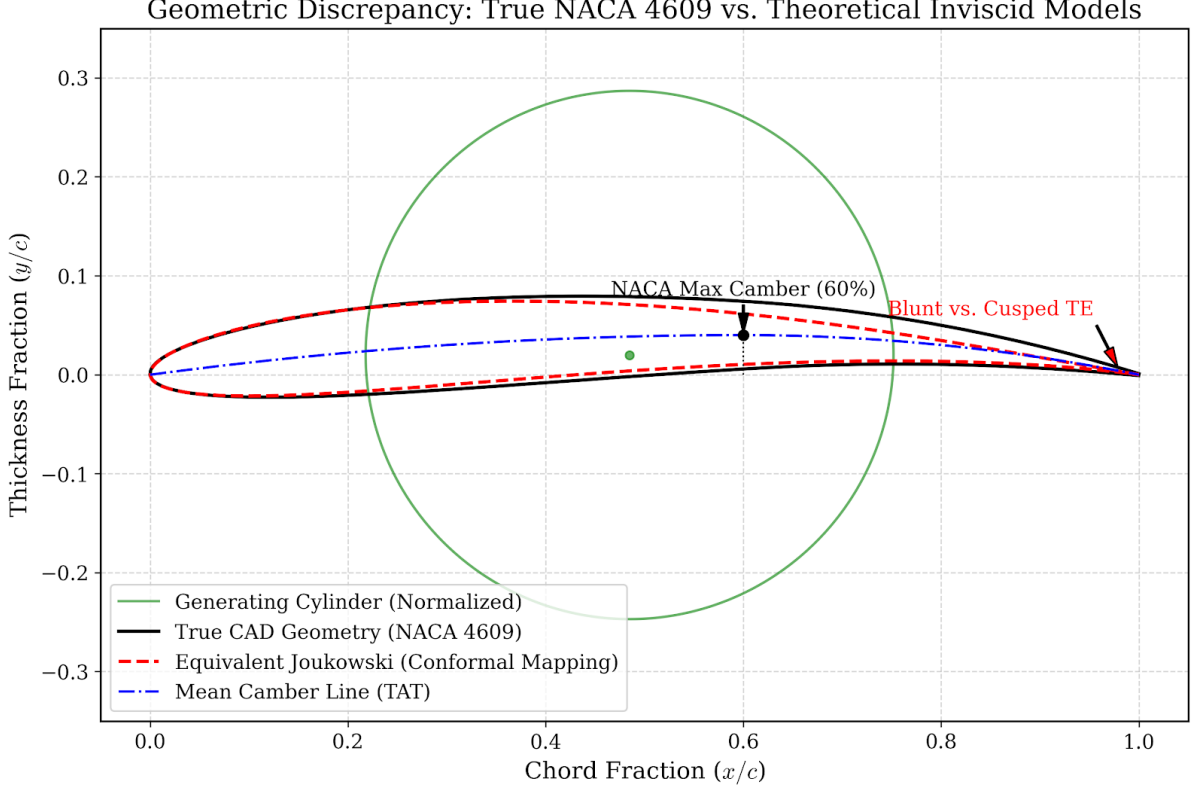


Figure 4: Geometric discrepancy between the true NACA 4609 CAD profile and the theoretical inviscid models generated by the MATLAB algorithm.

3 Numerical Methodology

To evaluate the true aerodynamic performance of the NACA 4609 taking into account the actual physics of flow including boundary layer dynamics, flow separation and viscous effects, a viscous Computational Fluid Dynamics (CFD) simulation was carried out. The simulations were performed using the finite-volume solver ANSYS Fluent in a steady-state, pressure-based and 2D setting.

3.1 Computational Domain and Mesh Generation

The physical/actual geometry of the NACA 4609 was defined and drawn with a unit chord of ($c = 1$ m) in SOLIDWORKS. The leading edge was placed exactly at the origin. To capture full aerodynamic effects and to minimize unfavorable domain wall effects, a C-grid far-field domain was drawn having the dimensions shown in figure 5. The relatively large domain ensures true flight characteristics. The global domain was discretized using unstructured triangular mesh since they provide greater flexibility especially for more complicated geometries (They also provide less manual input). An important requirement for full viscous simulations is the resolution of the boundary layer down to the viscous sublayer. A hybrid structured approach was used for the boundary layer. Furthermore, to strictly maintain a non-dimensional wall distance of $y^+ \leq 1.0$, the first cell height was calculated and explicitly set to 1.2×10^{-5} m. The boundary layer was resolved using 20 inflation layers extruded at a constant growth rate of 1.14.

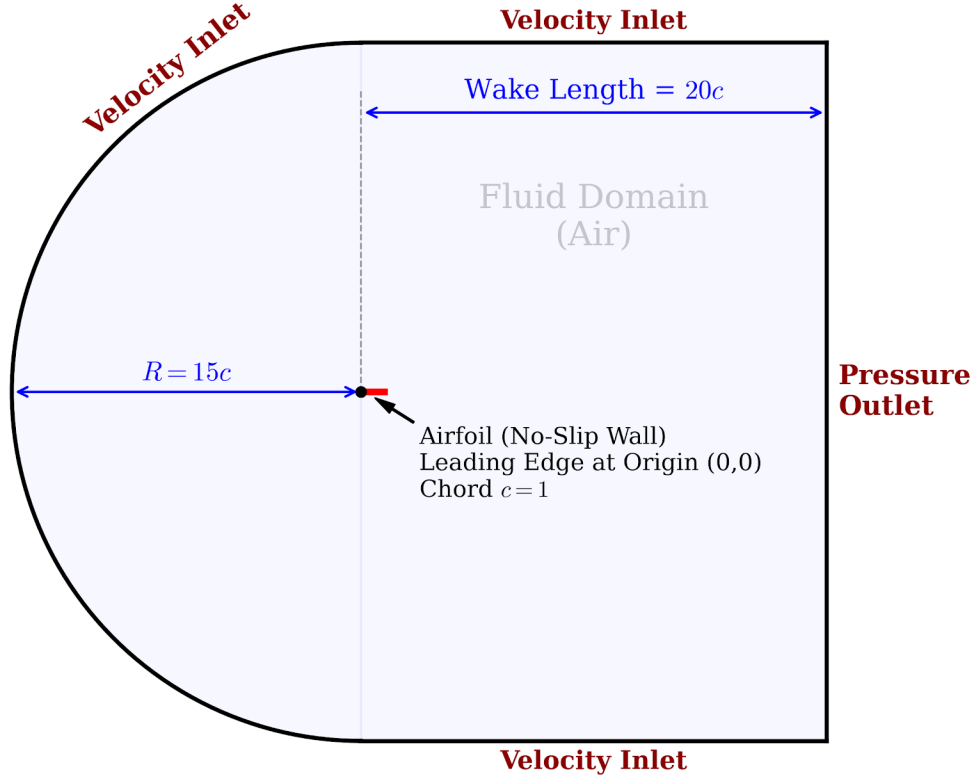
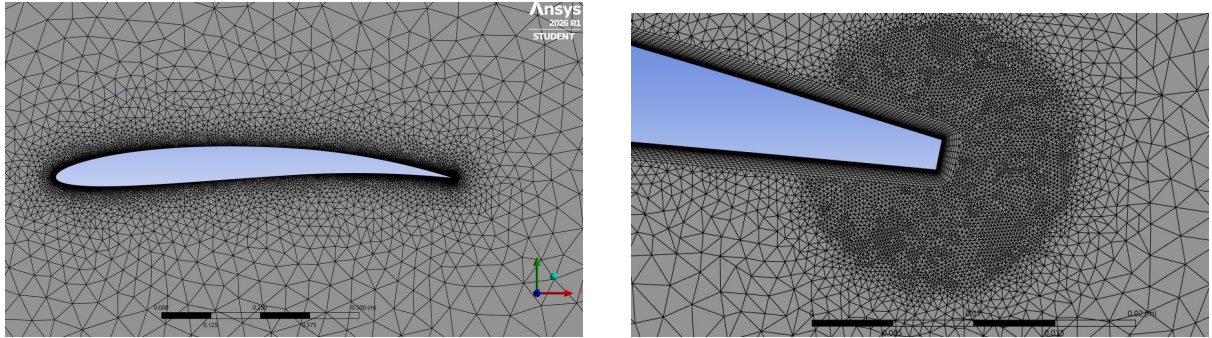


Figure 5: Schematic of the C-grid computational domain. The semi-circular inlet ($R = 15c$) ensures the freestream flow approaches undisturbed, while the extended $20c$ wake region allows for wake resolution behind the airfoil.



(a) Global C-grid topology.

(b) TE refinement and inflation layers.

Figure 6: Computational mesh detailing the discretization. The mesh smoothly transitions from the freestream domain to the viscous sublayer. The trailing edge in (b) was further refined using a sphere of influence coverage to better capture the physics and also reduce solution time.

3.2 Grid Independence Study

To ensure that the generated mesh/grid did not adversely affect the solution and the value of the force coefficients, a grid independence study was conducted. For this, a

baseline angle of attack of $\alpha = 5^\circ$ was chosen as the benchmark. Three distinct meshes were generated with the following metrics; A Coarse mesh having approx. 40,000 cells, A medium mesh with approx. 60,000 cells, and finally a fine mesh with approx. 106,000 cells. The logic for the mesh refinement followed a fixed factor of $r = 1.5$, where all mesh metrics except for the inflation layer (which was held fixed to ensure our desired $y+$ value) were parametrized and refined by r .

Table 1: Grid independence study results for the NACA 4609 at $\alpha = 5^\circ$, evaluating the sensitivity of aerodynamic force coefficients to spatial discretization.

Mesh Level	Elements	C_l	ΔC_l (%)	C_d	ΔC_d (%)	$C_{m,LE}$	$\Delta C_{m,LE}$ (%)
Coarse	$\approx 40,000$	1.059	–	0.0119	–	-0.399	–
Medium	$\approx 60,000$	1.048	1.03	0.0128	7.56	-0.394	1.25
Fine	$\approx 106,000$	1.048	0.00	0.0134	4.68	-0.393	0.25

The solution exhibited mesh independence, yielding a steady lift coefficient of $C_l = 1.048$ with a 0% change from the Medium mesh, while adequately resolving the pressure gradients required to compute drag ($C_d = 0.0134$) and pitching moment ($C_{m,LE} = -0.393$). Consequently, the 106k element mesh was selected for the full angle of attack sweep. The final grid demonstrated good topological health, maintaining an average Orthogonal Quality of 0.937 (minimum 0.47) and a stable average skewness of 0.13.

3.3 Physics Setup and Boundary Conditions

The freestream boundary condition was defined as velocity inlet with a fixed uniform velocity of $V_\infty = 30$ m/s across the entire angle of attack sweep. Moreover, to cover the planned angle of attack sweep with ease, only the incoming velocity vector was rotated to account for various angles of attack. The outlet was set as pressure outlet with a gauge pressure of $P_g = 0$ atm. The simulations were done at standard atmospheric conditions (density $\rho = 1.225$ kg/m³ and dynamic viscosity $\mu = 1.7895 \times 10^{-5}$ kg/m · s), the local Mach number was approximately $M \approx 0.1$, for which compressibility effects are considered to be negligible, justifying the adoption of strictly incompressible flow. Furthermore, The airfoil surface was designated as a no-slip wall.

Given our chord length (characteristics length) of 1, the Reynolds' number was calculated to be $Re \approx 2.05 \times 10^6$. At this Re , turbulence effects are significant and necessitates the use of a turbulence model; To accurately capture the onset of adverse pressure gradients and boundary layer detachment, the $k - \omega$ Shear Stress Transport (SST) turbulence model was selected. This two-equation formulation has become very effective for external aerodynamics with good capability to capture detachment and flow separation.

3.4 Numerical Schemes and Computation

The coupled pressure-velocity solver (utilizing a Rhie-Chow momentum-based scheme) was employed to accelerate convergence. Convergence was defined not only by a drop in scalar residuals below 10^{-5} , but also by monitoring the C_l , C_d , and C_m values vs iteration

until they reached a stable value and eventually flattened. All of the simulations, except for the one conducted at $\alpha = 20$ converged in under 200 iterations. The case with flow separation where $\alpha = 20$ ran to about 800 iterations before manually stopping it. As expected, this simulation would not converge under the assumption of a steady state since the complex wake dynamics would continue oscillating indefinitely, hence the failure to converge. As a result, Force coefficient values for this case were taken to be the average of these oscillations over approximately 300 hundred iteration and should be interpreted qualitatively.

The simulations were done on my personal laptop computer featuring an 11th gen intel core i7 11370H quad core CPU with 16 Gigabytes of RAM.

4 Results and Discussion

In order to better understand and visualize the limitations of inviscid theory (TAT and Conformal Mapping), the aerodynamic performance of the NACA 4609 was evaluated across an angle of attack sweep from $\alpha = 0^\circ$ to 20° . The full viscous CFD solution was plotted directly on top of the data predicted by our two inviscid theories.

4.1 Lift Characteristics and Stall Prediction

The lift coefficient (C_l) is plotted versus angle of attack (α) for all three methods in figure (7). In the region ($0^\circ \leq \alpha \leq 10^\circ$), all three methodologies seem to be in close agreement. This is to be expected since the flow is predominantly operating within the attached flow regime and viscous effects including boundary layer thickness are minimal. As a result, the inviscid methods still exhibit relevance even today for rapid estimation of aerodynamic force coefficients within that range. That said, as angle of attack α is increased beyond 10° , the full viscous solution starts diverging from the two inviscid methods, effectively capturing real-world physics behavior of boundary layer growth and the onset of flow detachment. The disparity starts growing even further as we approach 15° where the two inviscid models incorrectly predict a continually rising linear lift slope and completely ignoring stall conditions. As shown by the full viscous CFD solution, C_l experiences a sharp drop in the range beyond 15° .

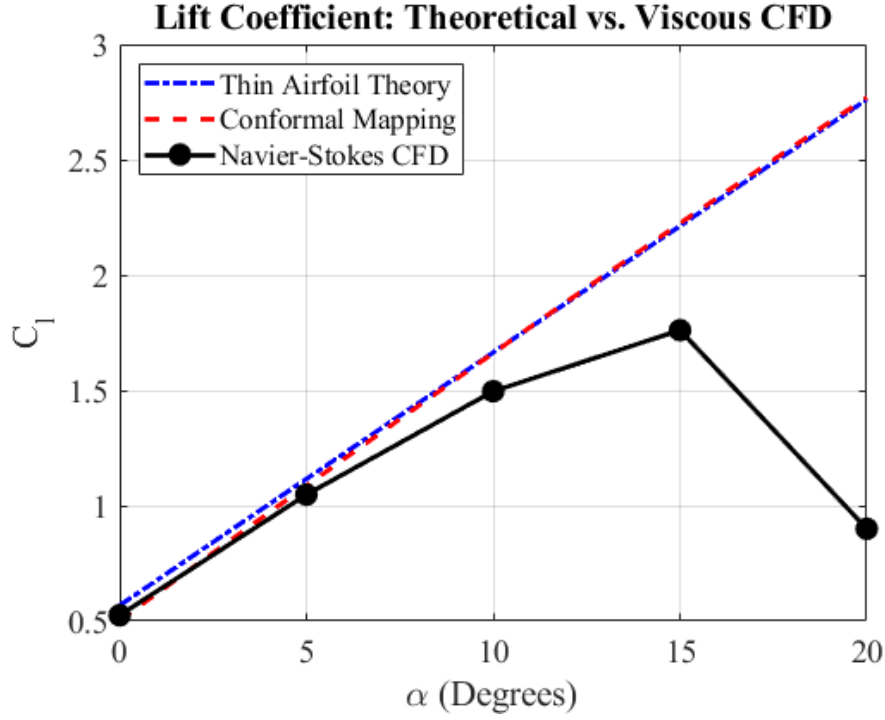


Figure 7: Comparison of lift coefficient (C_l) against angle of attack. The CFD successfully captures boundary layer detachment and deep stall at $\alpha = 20^\circ$, whereas the inviscid theoretical models incorrectly predict an infinitely rising lift slope.

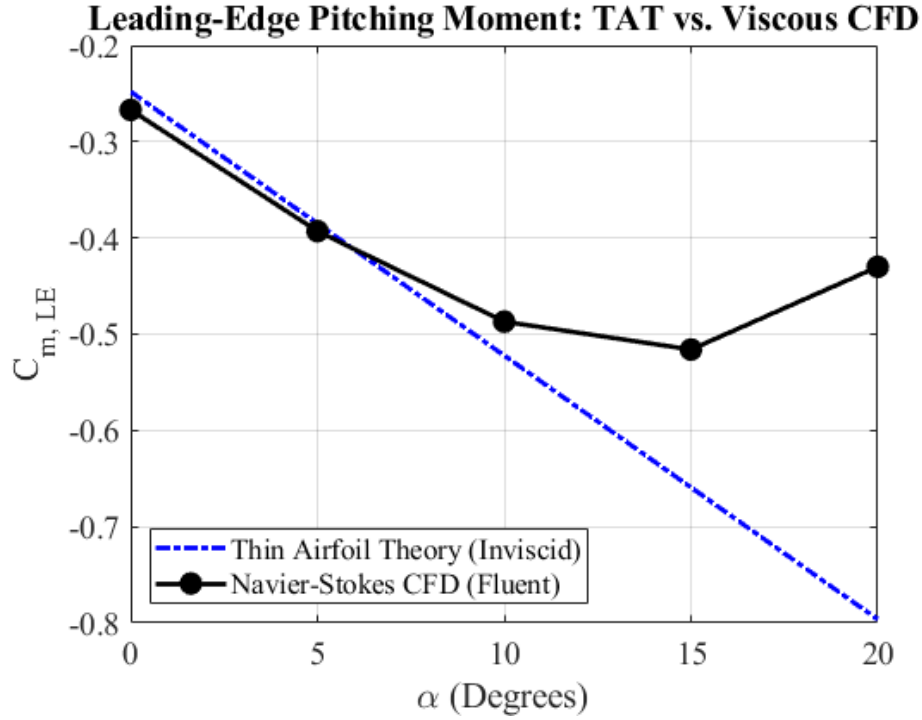


Figure 8: Comparison of the leading-edge pitching moment coefficient ($C_{m,LE}$) between Thin Airfoil Theory and viscous CFD. The inviscid TAT model predicts a strictly linear relationship, failing to capture the nonlinear shift in the center of pressure caused by boundary layer growth and flow separation at higher angles of attack.

As illustrated in Figure 8, a similar trend is visible in the moment coefficient graph. TAT seems to be in close agreement with the CFD solution in the predominantly attached flow regime, that is ($0^\circ \leq \alpha \leq 10^\circ$). Here the effects of friction and viscosity are minimal and the boundary layer very thin. As angle of attack α is increased further, we witness a divergence in the values predicted by either method. This is due to growth of the boundary layer thickness which alters the effective geometry of the airfoil that forces the center of pressure to shift, a phenomenon that inviscid models cannot capture. Inviscid models operate under the assumption of frictionless flow where the flow remains completely attached to the airfoil geometry. This graph further demonstrates the importance of Navier-Stokes CFD for capturing real-world physics.

4.1.1 Surface Pressure Distribution using TAT (C_p)

A main limitation of Thin Airfoil Theory (TAT), as we discussed, is the fact that it assumes the airfoil has zero thickness and is infinitely thin. As a result, the pressure distribution predicted by it will be symmetric across the camber line. Implying an equal and opposite C_p on the upper and lower surface of the airfoil. By modeling the airfoil as a singular vortex sheet of strength $\gamma(x)$, and applying the Kutta condition that $\Delta p = 0$ or equivalently $\gamma(x) = 0$ at the trailing edge, we will obtain the velocity:

$$u_{upper} = \frac{\gamma(x)}{2}, \quad u_{lower} = -\frac{\gamma(x)}{2} \quad (27)$$

Applying the linearized Bernoulli equation, the pressure coefficient (C_p) at any point along the surface becomes: (Notice how the term $(\frac{u}{V_\infty})^2$ vanishes as a consequence of $V_\infty \gg u$)

$$C_p = 1 - \left(\frac{V_\infty + u}{V_\infty} \right)^2 \approx -\frac{2u}{V_\infty} \quad (28)$$

Substituting the surface velocity perturbations yields the theoretical pressure coefficients for the upper and lower surfaces:

$$C_{p,upper} = -\frac{\gamma(x)}{V_\infty}, \quad C_{p,lower} = \frac{\gamma(x)}{V_\infty} \quad (29)$$

This result confirms that the pressure distribution is strictly a function of the vortex sheet since TAT inherently dismisses airfoil thickness. The result is a symmetric pressure distribution over the upper and lower surfaces of the airfoil which is physically incomplete and leaves a lot to be desired.

4.1.2 Surface Pressure Distribution using Conformal Mapping Technique (C_p)

To evaluate the pressure distribution over the Joukowski airfoil, we must first determine the velocity field in the physical z -plane. The tangential velocity on the surface of the complex circle in the ζ -plane is a superposition of the freestream flow and the induced circulation. For an arbitrary angle θ around the circle's center, this velocity is defined as:

$$V_\zeta(\theta) = 2V_\infty \sin(\theta - \alpha) + \frac{\Gamma}{2\pi R} \quad (30)$$

Substituting the previously derived circulation ($\Gamma = 4\pi R V_\infty \sin(\alpha + \beta)$), the velocity magnitude on the circle surface simplifies to:

$$V_\zeta(\theta) = 2V_\infty [\sin(\theta - \alpha) + \sin(\alpha + \beta)] \quad (31)$$

To map this velocity from the complex circle to the physical airfoil, we use the chain rule and take the derivative of the complex mapping function. The physical surface velocity, $V(z)$, then becomes the circle velocity scaled by the mapping modulus:

$$V(z) = \frac{|V_\zeta(\theta)|}{\left|\frac{dz}{d\zeta}\right|} = \frac{|V_\zeta(\theta)|}{\left|1 - \frac{a^2}{\zeta^2}\right|} \quad (32)$$

Finally, applying the incompressible Bernoulli equation, the theoretical pressure coefficient along the surface of the Joukowski airfoil is computed as:

$$C_p = 1 - \left(\frac{V(z)}{V_\infty}\right)^2 \quad (33)$$

Unlike TAT, the Conformal Mapping approach takes the airfoil thickness into account and provides a far more realistic pressure distribution (at lower angles of attack) over the airfoil. As it can accurately capture the leading edge stagnation point that yields a C_p value of 1.

4.2 Surface Pressure Distribution and Flow Physics

For our final comparison, pressure distributions were plotted at 3 angles of attack, mainly $\alpha = 0^\circ$, 5° and 15° using all three methods. For the full CFD solution, C_p data was extracted from ANSYS Fluent and plotted accordingly.

4.2.1 Attached Flow Regime ($\alpha = 0^\circ$ and 5°)

At $\alpha = 0^\circ$ and $\alpha = 5^\circ$, the pressure distributions signify predominantly attached flow with no separation. Moreover, there is a remarkably close agreement between the Conformal Mapping approach and the full CFD solution. This stems from the fact that the assumption of thickness (9% for NACA 4609) is baked into the Joukowski Conformal Mapping technique, whereas Thin Airfoil Theory (TAT) assumes an infinitely thin airfoil that yields a non-physical symmetric pressure distribution.

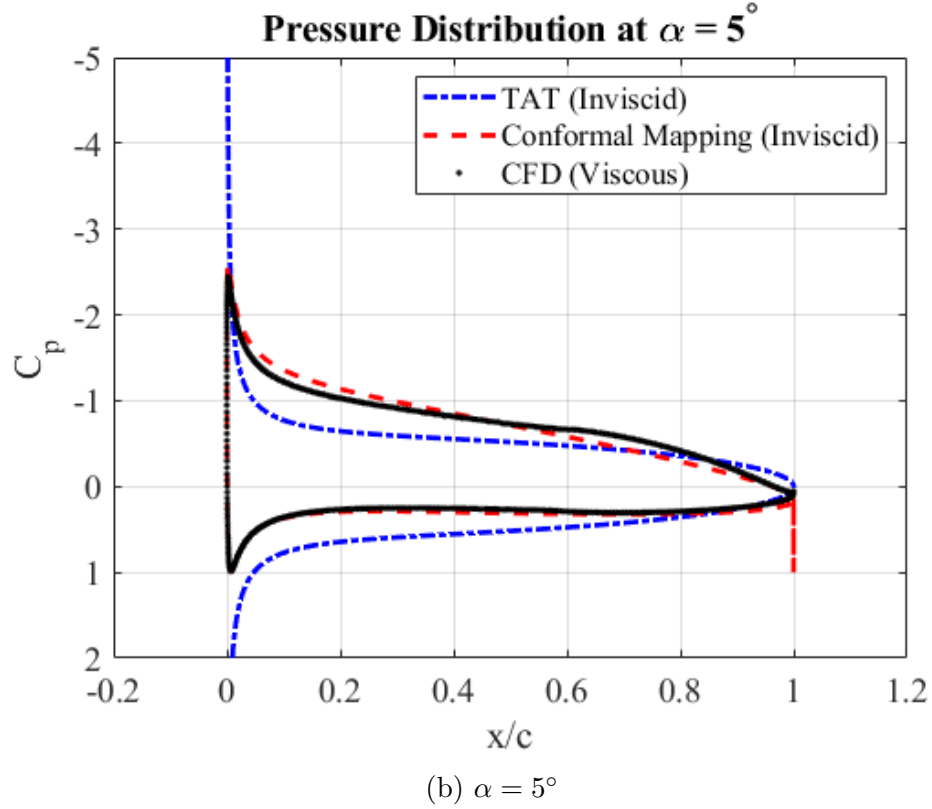
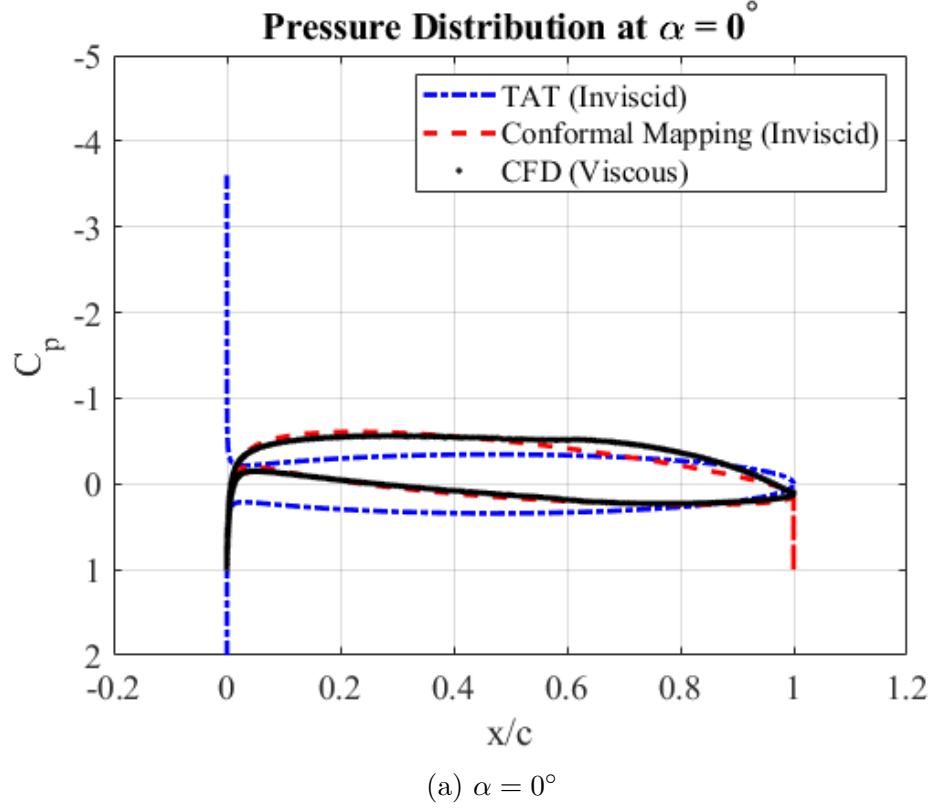


Figure 9: Surface pressure coefficient (C_p) distribution comparing inviscid analytical theories and viscous CFD across the predominantly attached flow regime at $\alpha = 0^\circ$ and 5° .

4.2.2 Boundary Layer Separation and Onset of Stall ($\alpha = 15^\circ$)

At $\alpha = 15^\circ$, the pressure distribution reveals the physical limitations of inviscid models. Since TAT and Conformal Mapping assume ideal, frictionless flow, they continue to predict rising suction peaks without accounting for adverse pressure gradients near the trailing edge. Whereas this phenomenon is well accounted for in the Fluent CFD solution, where we witness a flattening of C_p near the trailing-edge, specifically in the region from $\frac{x}{c} = 0.8$ to 1.

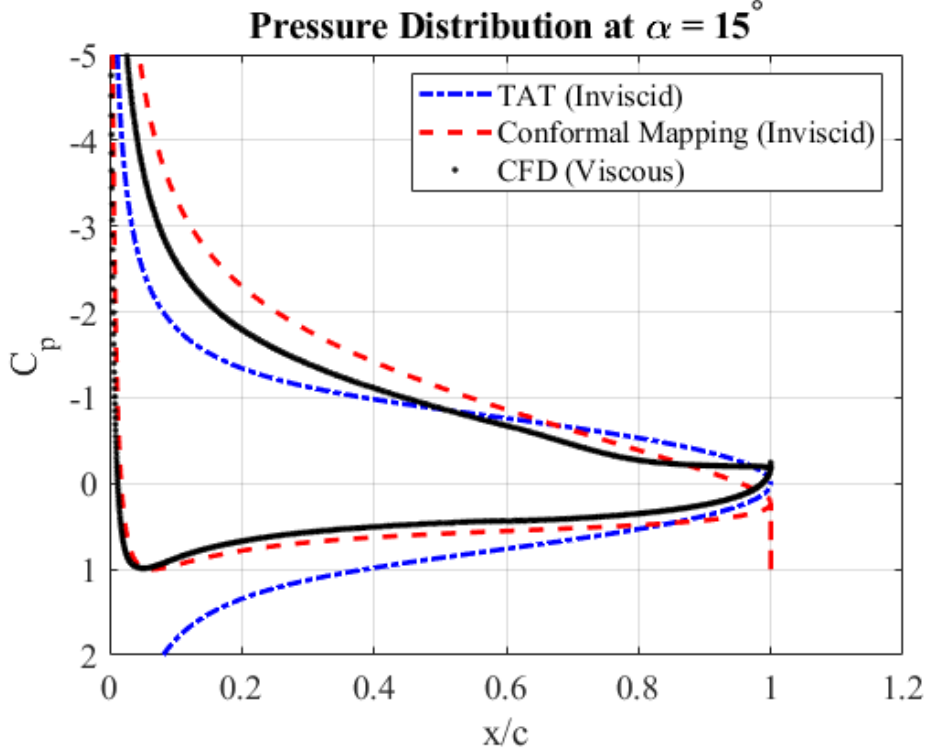
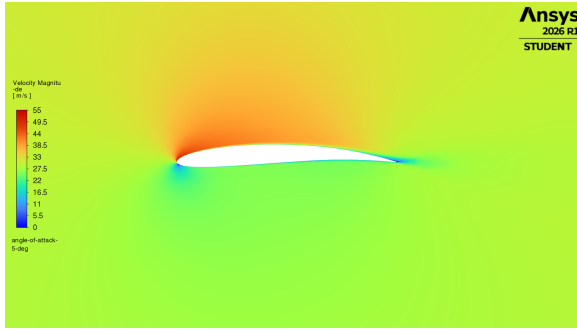


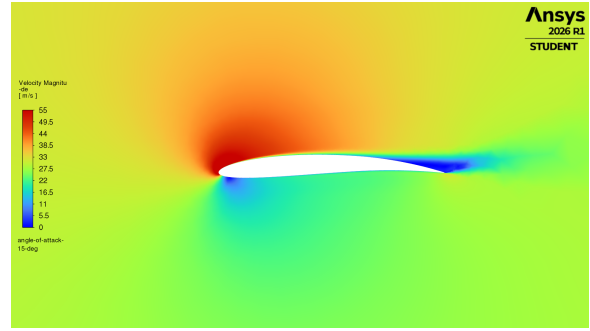
Figure 10: Surface pressure coefficient (C_p) distribution at $\alpha = 15^\circ$, capturing the pressure plateau indicative of trailing-edge boundary layer separation.

4.2.3 Flow Field Contours

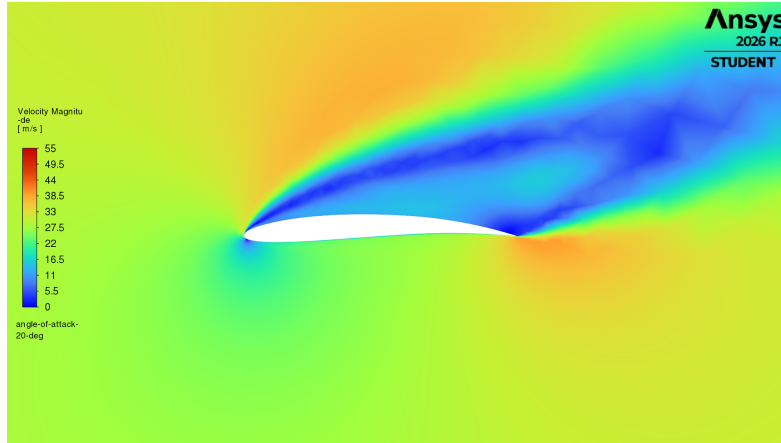
In order to verify the results, and better visualize the physics of flow, the velocity contours are presented below. At lower angles of attack ($\alpha = 5^\circ$), the flow smoothly accelerates over the upper surface. Stagnation point at the leading edge is also clearly visible. At $\alpha = 15^\circ$ and subsequently into deep stall at $\alpha = 20^\circ$ (where steady-state residuals failed to converge due to large-scale, unsteady vortex shedding dynamics), the velocity contours illustrate a massive flow separation indicative of stall and flow reversal leading to a sudden and sharp decrease in lift. This visualization confirms that, even though classical inviscid theories are very elegant mathematically, they fall short of full viscous CFD solution. For full Navier-Stokes' equations are an indispensable tool for capturing real world aerodynamic performance and for determining the true operational limits and stall behavior of an airfoil.



(a) Attached boundary layer ($\alpha = 5^\circ$)



(b) Onset of separation ($\alpha = 15^\circ$)



(c) Deep stall and flow detachment ($\alpha = 20^\circ$)

Figure 11: Normalized ANSYS Fluent velocity magnitude contours showing the progression of boundary layer separation. At 20° , the flow completely separates and becomes highly transient, reflecting the physical reality behind the solver's steady-state non-convergence.

5 Conclusion

An objective was set for this study to compare three different approaches to airfoil analysis. Two classical inviscid models; Thin Airfoil Theory and the Joukowski Conformal Mapping technique were pitted against a 2D viscous Navier-Stokes CFD simulation. Baseline data was generated using a custom MATLAB solver for each inviscid approach, and the algorithm was generalized to output force coefficients for any arbitrary NACA 4-digit airfoil. The two inviscid approaches demonstrated excellent agreement with the viscous CFD solver in the predominantly attached flow region ($\alpha = 0^\circ$ to 10°), where viscous effects were minimal. Moreover, the Conformal Mapping approach showed surprisingly accurate pressure-distribution predictions, as it accounts for airfoil thickness.

That said, where inviscid models fall short is in the more extreme flow scenarios where boundary layer thickness and flow separation become significant and their dismissal causes large discrepancies in their predicted values. These discrepancies became visible once we exceeded $\alpha = 10^\circ$, and peaked at $\alpha > 15^\circ$.

Inviscid theories completely ignore the effects of viscosity, and this is apparent in the infinitely rising linear lift slope that they predict. Furthermore, they fail to accurately

capture the adverse pressure gradients in the trailing edge of the airfoil where the viscous CFD solution clearly illustrates a plateau in the C_p plot on the upper surface of the airfoil.

However, due to their ultra low computational costs and rapid solution generation, inviscid methods still remain an excellent tool for quick workflows and rapid design explorations. Granted we do not exceed their limits and remain within the range where these theories perform best.

In conclusion, the viscous Navier-Stokes CFD solvers remain as the go-to tool for fully exposing the aerodynamic limitation of any lifting body and are invaluable for fully capturing real-world physics of flow.

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